although we have tied our discussion to one particular means for determining the electron's whereabouts, Heisenberg showed that the trade-off between the precision of position and velocity measurements is a fundamental fact that holds true regardless of the equipment used or the procedure employed. Unlike the framework of Newton or even of Einstein, in which the motion of a particle is described by giving its location and its velocity, quantum mechanics shows that at a microscopic level you cannot possibly know both of these features with total precision. Moreover, the more precisely you know one, the less precisely you know the other. And although we have described this for electrons, the ideas directly apply to all constituents of nature.

Einstein tried to minimize this departure from classical physics by arguing that although quantum reasoning certainly does appear to limit one's knowledge of the position and velocity, the electron still has a definite position and velocity exactly as we have always thought. But during the last couple of decades theoretical progress spearheaded by the late Irish physicist John Bell and the experimental results of Alain Aspect and his collaborators have shown convincingly that Einstein was wrong. Electrons—and everything else for that matter—cannot be described as simultaneously being at such-and-such location and having such-and-such speed. Quantum mechanics shows that not only could such a statement never be experimentally verified—as explained above—but it directly contradicts other, more recently established experimental results.

In fact, if you were to capture a single electron in a big, solid box and then slowly crush the sides to pinpoint its position with ever greater precision, you would find the electron getting more and more frantic. Almost as if it were overcome with claustrophobia, the electron will go increasingly haywire—bouncing off of the walls of the box with increasingly frenetic and unpredictable speed. Nature does not allow its constituents to be cornered. In the H-Bar, where we imagine ħ to be much larger than in the real world, thereby making everyday objects directly subject to quantum effects, the ice cubes in George's and Gracie's drinks frantically rattle around as they too suffer from quantum claustrophobia. Although the H-Bar is a fantasyland—in reality, ħ is terribly small—precisely this kind of quantum claustrophobia is a pervasive feature of the microscopic realm. The